

Project Title
Digital Library Search System

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Describe what your project presents within 150 words.

The **Digital Library Search System** is an algorithm-driven information retrieval platform designed to help users efficiently search and discover books, journals, magazines, and research papers from a large digital collection. The system goes beyond conventional keyword matching by applying advanced string-processing and text-analysis techniques to improve **search speed, accuracy, and relevance**.

Users can search resources using titles, authors, ISBNs, categories, and keywords. The system implements **Knuth-Morris-Pratt (KMP), Rabin-Karp, and Boyer-Moore algorithms** for efficient pattern searching across library content. **Edit Distance** enables fuzzy searching, allowing relevant resources to be retrieved even when search queries contain spelling errors or minor variations. **Document similarity** is used to identify related resources, while **citation-flow analysis** helps represent connections between research papers and supports related-resource discovery. **Suffix-array-based searching** can further support efficient processing of large text collections.

The system will also analyze and compare the **time and space complexity** of different algorithms to determine their suitability for different search operations. Selected graph, optimization, randomized, and parallel algorithmic techniques from the project syllabus will be demonstrated through suitable library-oriented problem scenarios.

The final system aims to provide a **fast, flexible, and intelligent digital search experience**, demonstrating how appropriate algorithm selection can transform a basic library repository into an efficient information-retrieval system.

Module-Wise Implementation:

Module 1: Pattern search, fuzzy matching, document similarity, citation-flow analysis, and text-based query processing.

Module 2: KMP, Rabin-Karp, Boyer-Moore, hashing, and suffix-array-based searching.

Module 3: Dynamic programming techniques for Edit Distance and related text-processing operations.

Module 4: Graph-based citation and resource-relationship analysis.

Module 5: Selected optimization and approximation problem demonstrations.

Module 6: Randomized and parallel processing demonstrations for large library datasets.

Signature of the Guide